Composites: Part A 207 (2026) 109863

Contents lists available at ScienceDirect

Composites Part A

journal homepage: www.elsevier.com/locate/compositesa

Interpretable prediction of ultimate strength of composite pressure hulls using data-driven symbolic regression

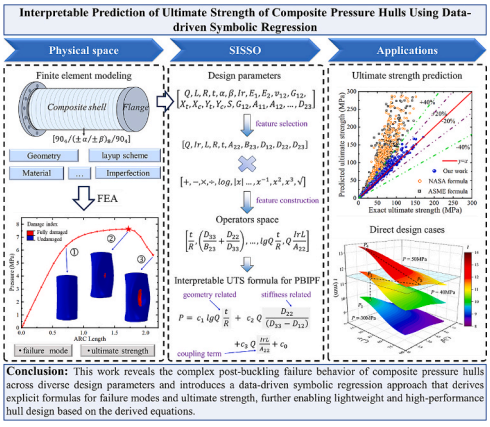
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HIGHLIGHTS

- Failure modes and ultimate strength of composite pressure hull are predicted.
- Explicit, interpretable formulas are derived via data-driven symbolic regression.
- The formulas achieve accuracy and outperform existing empirical models.
- Direct design strategy for lightweight, high-performance hulls are demonstrated.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:
Composite pressure hulls
Ultimate strength prediction
Data-driven symbolic regression
Explicit formula

ABSTRACT

Pressure hulls of submersibles are typically manufactured from multilayered composite cylindrical shells to withstand deep-sea hydrostatic pressure. Accurate prediction of their ultimate strength (UTS) remains challenging due to complex post-buckling behavior and strong coupling among multiple design parameters. This work presents an interpretable data-driven approach that establishes universal expressions for predicting the UTS of composite pressure hulls. Experimentally calibrated finite element analysis (FEA) is first performed to build a comprehensive failure database covering various design parameters. Then, feature engineering and symbolic regression are employed to extract critical descriptors and derive explicit UTS formulas. The formulas are validated against published data, with errors below 13%. Furthermore, comparisons between the derived and existing empirical formulas, such as ASME 2007 and NASA SP-8700, based on our FEA database demonstrate that the proposed formulas deliver markedly superior performance. Unlike the black box nature of conventional AI-based models, the derived formulas possess clear physical interpretability, directly revealing the relationships between design variables and the corresponding UTS. This enables direct design of lightweight or high-performance composite pressure hulls through rational parameter selection. As a result, this work provides a

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https://doi.org/10.1016/j.compositesa.2026.109863
Received 3 November 2025; Received in revised form 2 April 2026; Accepted 16 April 2026
Available online 17 April 2026
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practical strategy for developing data-enhanced predictive models to support the design and failure evaluation of composite hulls.

1. Introduction

Composite cylindrical shells are widely adopted as submersible pressure hulls owing to their high specific strength, low weight, and excellent corrosion resistance [1,2], and thus play an essential role in ocean engineering. As diving depth increases, the required shell thickness rises accordingly to resist greater hydrostatic pressure. However, the failure mechanisms of thick composite hulls are more complex than those of thin-walled or metallic hulls due to the strong interaction between structural buckling and progressive material failure [3,4]. This complexity makes accurate prediction of their ultimate strength (UTS)—a crucial index to ensure structural safety—particularly challenging. Moreover, the UTS of composite hulls is highly sensitive to multiple, coupled design variables [5,6], while reliable prediction methods remain limited, thereby constraining efficient design and optimization.

Extensive efforts have investigated buckling and post-buckling behaviors of thin composite hulls under external pressure, using eigenvalue and nonlinear analyses as well as analytical approximations to estimate critical loads [7–9]. However, with increasing thickness, approaches developed for thin hulls become insufficient for collapse prediction because they typically neglect the stiffness degradation from material damage, which may trigger premature instability [10,11]. Jiang et al. [11] performed hydrostatic tests on moderately thick composite hulls, reporting that failure was dominated by material damage. Cong et al. [12] demonstrated via numerical simulations that the failure mode transitions from circumferential buckling to strength failure and axial buckling as geometric parameters vary. Overall, for thicker composite hulls, failure mode and UTS are controlled by the coupled effects of structural buckling and material failure [12–14], which remain not fully understood.

The broad design freedom of composites further enlarges the parameter space of composite hulls (i.e., geometry, material, lay-up, imperfection amplitude). UTS under hydrostatic pressure is highly sensitive to these design parameters [14,15], making quantitative, comprehensive characterization crucial for rational design. However, deep-sea testing is costly and high-fidelity nonlinear FEAs are time-consuming, so many studies are restricted to specific layup schemes or geometric configurations, with few encompassing broader design variables [16,17]. The high dimensionality and nonlinearity between UTS and design variables, together with imperfection sensitivity [18], pose additional challenges for theory-driven prediction across a wide design space. These limitations motivate new approaches that can deliver accurate UTS prediction over broad design configurations.

Machine learning (ML) has shown promise for accelerating composite structural design [19,20]. Tahir et al. [21] employed an artificial neural network (ANN) to predict the buckling load of thin cylindrical shells under axial compression. Liu et al. [22] developed a surrogate model by integrating FEA with response surface method to optimize the stiffening rib configuration of composite hulls. However, conventional ML models often behave as black boxes and lack explicit, physically interpretable formulas required in engineering applications [23–25]. They may also be tailored to specific domains and struggle to generalize [26,27]. Symbolic regression (SR) addresses these limitations by discovering analytical equations directly from data. Among SR approaches, genetic programming (GP) has been applied to develop continuum damage models for composites[28] and to predict impact resistance[29], though it can be computationally expensive and produce less interpretable formulas[30]. In contrast, the Sure Independence Screening and Sparsifying Operator (SISSO) algorithm [31–33] offers a powerful route to extract low-dimensional descriptors from high-dimensional features and to derive concise formulas with strong predictive ability. Tian et al. [32] successfully classified materials with positive or negative thermal expansion using SISSO-derived descriptors. Pan et al. [33] developed reliable SISSO-based equations to predict diffusion coefficients in covalent organic frameworks. Recent studies have also extended SISSO to composites, including the screening of flame-retardant polymeric composites[34] and the design of hybrid composites with zero-temperature coefficient of resistance[35]. These studies demonstrate the SISSO's capability in addressing multidimensional design-performance mapping problems. Thus, it is anticipated that SISSO can yield explicit formulas for predicting the UTS of composite hulls within high-dimensional design spaces.

As a result, we integrate FEA with SISSO to develop explicit formulas that cover a broad range of design parameters for composite hulls, enabling the determination of their failure modes and the prediction of their UTS. First, a comprehensive database that captures post-buckling and progressive damage across geometry, lay-up, material properties, and imperfections is established based on experimentally calibrated FEA. The influence of key design variables on the failure modes and UTS is systematically characterized through parametric, correlation, and importance analyses. Subsequently, the SISSO algorithm is employed on this database to extract explicit expressions that: (i) classify two typical failure modes—post-buckling-induced progressive failure (PBIPF) and material-damage-dominated failure (MDDF), and (ii) derive explicit UTS formulas for each mode. The derived expressions are validated against our FEA data and published experimental data, and are also compared with existing empirical equations. Results demonstrate the superior predictive performance of the derived formulas. Moreover, the formulas are highly interpretable, explicitly mapping the relationship between design parameters and UTS, which enables their direct application in the lightweight design and strength assessment of composite pressure hulls.

2. Model and methodology

Fig. 1 presents the workflow for deriving UTS prediction formulas of composite hulls and subsequently utilizing the formulas for direct design. The process initiates in the physical design space, where FEA of composite hulls is performed to investigate their failure mechanisms under varying design parameters. The effects of these parameters on both the failure mode and UTS are then systematically assessed, leading to the establishment of a failure analysis database. Building on this database, the SISSO method is employed to develop surrogate models, which enable classification of failure modes and prediction of UTS. This step yields explicit expressions for the UTS of composite hulls. Finally, the predictive performance of the obtained formulas is verified through comparison with published works and existing empirical equations, and several direct design cases are provided to demonstrate the utility of these interpretable formulas in the lightweight or high-performance design of composite hulls.

2.1. Finite element model and design parameters of composite hulls

The composite hull consists of a metal flange and a composite cylindrical shell, as shown in Fig. 2. The right flange is tied to the shell, while the left end is fully constrained [6]. The flange is made of 7075 aluminum alloy with modulus $E = 71.7$ GPa, Poisson's ratio $\nu = 0.33$, and thickness $t_m = 25$ mm. The composite shell has a radius R, length L, thickness t, and the layup scheme $[90_4/(\pm\alpha/\pm\beta)_8/90_4]$. The ranges of these design parameters are summarized in Table 1. C3D8R and S4R elements are used to discretize the flange and composite shell [8,11,36], respectively. The element size is selected as 8 mm, which is determined through convergence analysis. External hydrostatic pressure is applied

to the shell surface. To evaluate the influence of material properties on the failure mode and UTS, three composites from the literature are analyzed, with their properties listed in Table 2. All simulations are conducted using ABAQUS.

The overall failure analysis procedure for the composite hulls is depicted in Fig. S1. The composite hulls are initially established within the physical design space. Linear buckling analysis is performed to obtain the eigenvalue Q and the corresponding first-order mode. Based on the modal defect method, the first-order linear buckling mode of the hull is applied as a geometric imperfection to the ideal hull model. The imperfection amplitude of this mode is defined as $Ir \times R$, where Ir is imperfection ratio, considered as a design variable in Table 1. In accounting for non–idealized shapes, Ir is defined as:$\frac{w_{max}}{R}$ to achieve a reasonable assessment of strength, where w_{max} is the maximum deflection amplitude. With this foundation, a nonlinear buckling analysis incorporating material failure is carried out to calculate the UTS of composite hulls. Damage initiation is evaluated using the Hashin criterion, and the Chang stiffness degradation method is implemented via a USDFLD subroutine [11] to update the material stiffness accordingly. Once the stress state $(\sigma_{11}, \sigma_{22}, \sigma_{12})$ in a composite layer satisfies the Hashin failure criterion, as shown in Eqs. (S1-S5), the layer undergoes stiffness degradation. The Chang method utilizes specific reduction factors for different failure modes to model material stiffness degradation, as provided in Table S1. The above process is iteratively repeated until the hull fails completely, and eventually the failure mode and the UTS are obtained. Notably, the present study models geometric imperfections exclusively using the first-order linear buckling mode, which may not capture the full spectrum of manufacturing-induced defect patterns. Higher-order, mixed-mode, and stochastic imperfections may lead to different mechanical behaviors of pressure-resistant shells, which are not considered in the present work.

2.2. SISSO method

SISSO is a compression-aware data analysis method that combines sure independence screening (SIS) with sparse operators (SO) to identify low-dimensional descriptors that exhibit the strongest correlation with target properties, as shown in Fig. 3. The descriptor in this work is specifically composed of the design parameters of the composite hull. The SISSO procedure involves two main steps: feature space construction and descriptor identification. In the first step, SIS constructs an extensive feature space by iteratively applying a predefined set of algebraic operators to the original features. In the second step, SO evaluates all possible combinations of the SIS-generated features under regularization constraints, selecting the most relevant and parsimonious descriptors. Through this process, SISSO effectively extracts optimal low-dimensional descriptors from high-dimensional and highly nonlinear data, enabling an accurate and interpretable representation of the specified property. In this work, the maximum feature complexity and descriptor dimension are both limited to 3 to ensure model simplicity and generalizability. The feature construction is constructed using the following operator set: [(+)(−)(*)(/)(exp)(exp-)({\,}^-1)({\,}^2)({\,}^3)(sqrt)(cbrt)(log)(|-|)(scd)({\,}^6)(sin)(cos)].

Both classification and regression SISSO models are developed in this work. For the classification model, accuracy, precision, and recall, as defined in Eqs. (S6-S8), are utilized as evaluation metrics. For the regression model, the prediction accuracy is evaluated by root-mean-square-error (RMSE) and the goodness of fit R^2, as given in Eqs. (S9-S10).

2.3. Data collection for SISSO

In this section, two design spaces are addressed to enhance the

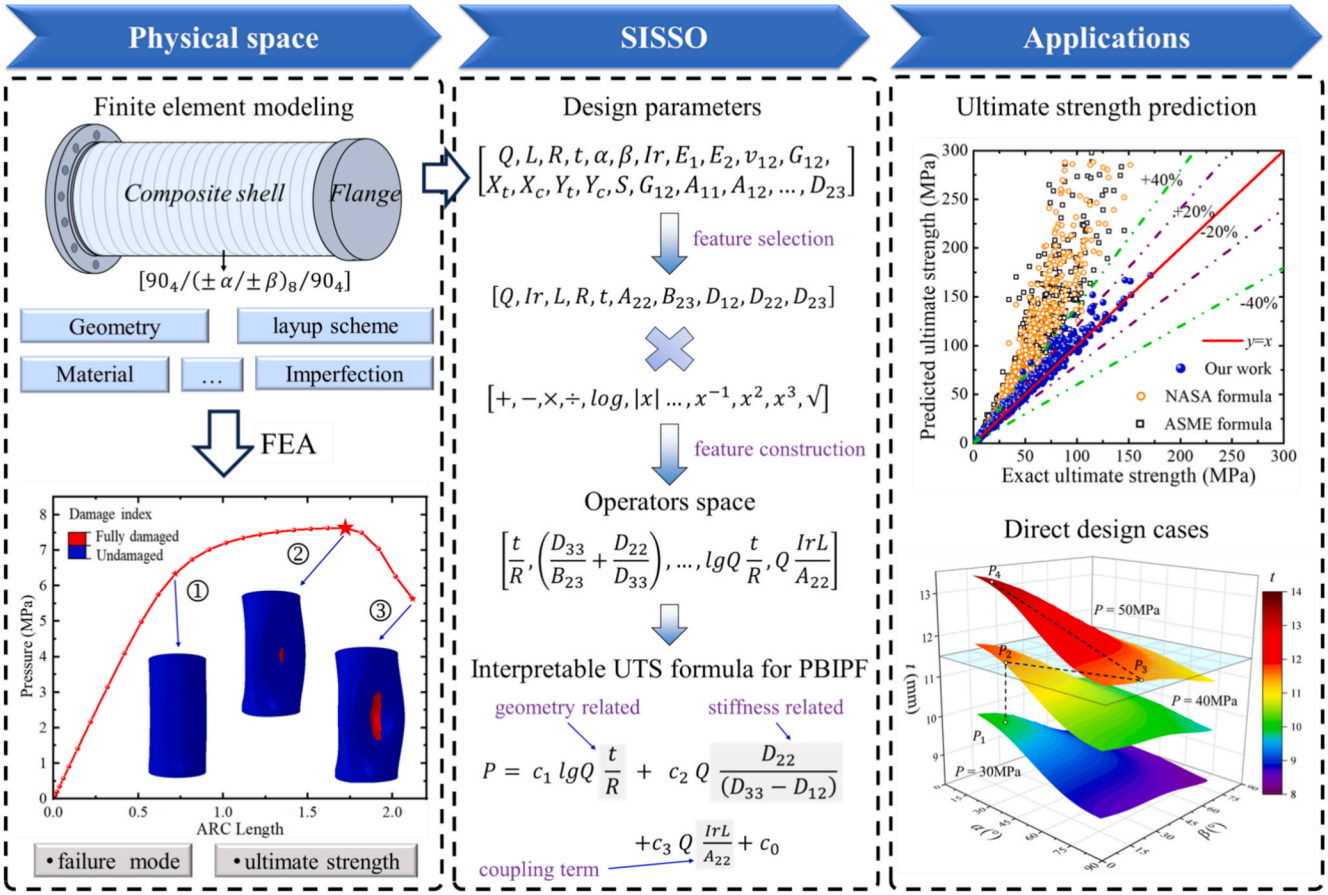


Fig. 1. The integrated workflow for deriving explicit formulas for the UTS of composite hulls and subsequently utilizing the derived formulas for direct design.

generalizability of the SISSO model, as summarized in Table 3. For design space 1, 1000 samples are randomly generated. Specifically, the material is T700/Epoxy, the range of length L is [300,800] mm, the radius R is within [80,180] mm, the layup scheme is $[90_4/(\pm\alpha/\pm\beta)_8/90_4]$ and the angles of layer α, β are both within [10°, 80°]. In design space 2, two new materials, i.e., T800G/5228 and M40J/TDE-85, have been introduced, with the imperfection ratio *Ir* of 0.8%, as referenced to [11], while other design parameters remain unchanged as given in design space 1. For each material, 1000 samples are randomly generated. Design space 2 also consists of 1000 samples with a new layup scheme $[(\pm\alpha/\pm\beta)_{10}]$. It is noteworthy that the number of samples in each design space is determined to balance prediction accuracy and computational cost of data generation.

3. Results and discussions of finite element analysis

3.1. Post-buckling behavior of composite hulls

The reliability of FEA is first verified by experiments. Chen et al. [6] and Jiang et al. [11] conducted hydrostatic tests on composite shells to identify the UTS. Results are listed in Table 4. The composite shell in [6] was made of T700 composite material with a length of 385 mm, radius of 100 mm, thickness of 5 mm, and the layup scheme $[90_4/(90/\pm45/0)_8/90_4]$, while the composite shell in [11] consisted of T700/Epoxy resin composite, with a length of 466 mm, radius of 67.5 mm, thickness of 12 mm, and the layup $[(90_2/\pm20/90_2/\pm45/90_2/\pm70)_5/90_4]$. By contrast, the failure modes and UTSs of the two composite shells are calculated with the simulation strategy given in Section 2.1. Results are given in Table 4. The UTSs calculated through FEA differ from the experimental values by less than 6%, demonstrating the reliability of our simulations in determining the UTS of composite hulls.

We further compare the progressive failure process of the experiment in reference [6] using our simulation. Fig. 4(a) presents the corresponding load versus arc length curve. At point 1, the composite hull undergoes critical buckling under a hydrostatic pressure of 6.34 MPa, leading to a reduction in structural stiffness, though without immediate loss of load-bearing capacity. As illustrated in Fig. 4(b), the material remains undamaged at this stage and can fully recover its original shape upon unloading. With continued loading, the structure enters a nonlinear post-buckling stage. The curve reaches its peak at point 2, where the hull collapses under 7.63 MPa, accompanied by pronounced deformation. As shown in Fig. 4(b), localized dents combined with fiber failures appear in the outer mid-section of the cylindrical segment. By point 3, fiber and shear failure regions expand rapidly, resulting in a complete loss of load-carrying capacity—consistent with experimental findings reported in [6]. Additionally, interactions between the end caps and fixed boundary conditions induce stress concentrations near the edges, causing matrix failure at both ends of the hull. A similar failure behavior is observed in the composite shell studied in reference [12,37]. We therefore define this type of failure—initiated by significant buckling, followed by fiber-dominated material damage, and characterized by the coupling of both mechanisms in the post-buckling phase—as PBIPF [38]. Using the current finite element framework, which was also adopted in [4,6,8], we successfully reproduce the complex post-buckling response and failure modes of literature models. These results confirm that our simulation can effectively capture the hydrostatic buckling and progressive damage behavior of composite hulls, thereby providing a reliable basis for data collection in this study.

Our extensive FEA results also revealed another failure mode for the composite hull. Fig. 4(c-d) shows the load versus arc length curve and the progressive failure process of the composite hull, which has the layup $[90_4/(\pm71/\pm69)_8/90_4]$ and is fabricated from T700/Epoxy with dimensions of 800 mm in length, 120 mm in radius, and 15 mm in thickness. When the composite hull is subjected to a hydrostatic pressure of 23 MPa at point 1, matrix damage occurs without obvious buckling, resulting in reduced stiffness. As the load increases, material failure intensifies. The hull eventually collapses at point 2 under a hydrostatic pressure of 37.52 MPa. At this stage, matrix failure develops at both ends and midspan of the cylinder, while the fibers remain intact. The matrix damage at the midspan is attributed to stress concentrations induced by localized indentations, whereas the failures at the ends are associated with boundary-induced stress concentrations. Overall, the structure failure is primarily caused by extensive matrix failure. We define this failure mode, which lacks significant buckling, as MDDF. Both the UTS and the failure modes (PBIPF and MDDF) are closely tied to the design parameters of composite hulls. Hence, it is essential to clarify and establish a characterization between these design parameters and structural performance to enable the rational design of composite hulls.

Table 1
Design parameters of composite hulls.

Design parameter	Interval
Radius R / mm	[80,180]
Length L / mm	[300,800]
Thickness t / mm	[5,20]
Angle of layer α,β	[10°, 80°]
Imperfection ratio *Ir*	[0.1%, 1%]

Table 2
Material properties of three composites. (MPa for modulus and strength).

Material Property	T700/ Epoxy	T800G/ 5228	M40J/TDE-85
Elastic modulus in fiber direction E_1	102,000	147,000	218,000
Elastic modulus in transverse direction E_2	7000	9320	9070
Poisson's ratio ν_{12}	0.16	0.3	0.3
Shear modulus G_{12}	3350	4530	6350
Longitudinal tensile strength X_t	1400	2820	2088
Longitudinal compressive strength X_c	1050	1538	1212
Transverse tensile strength Y_t	28	60.2	22.5
Transverse compressive strength Y_c	105	190	196
Shear strength S	75	125	61.1

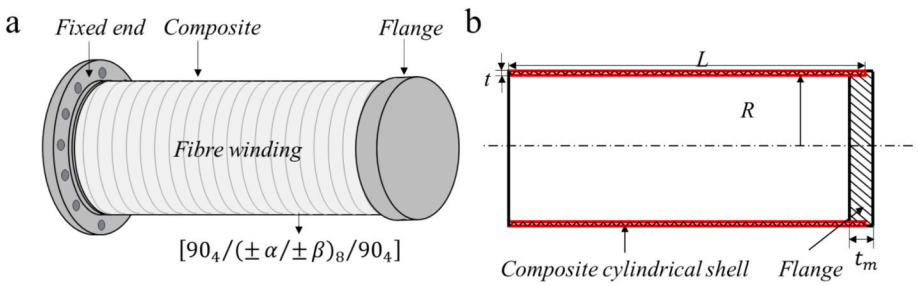


Fig. 2. Schematic diagram of composite hulls.

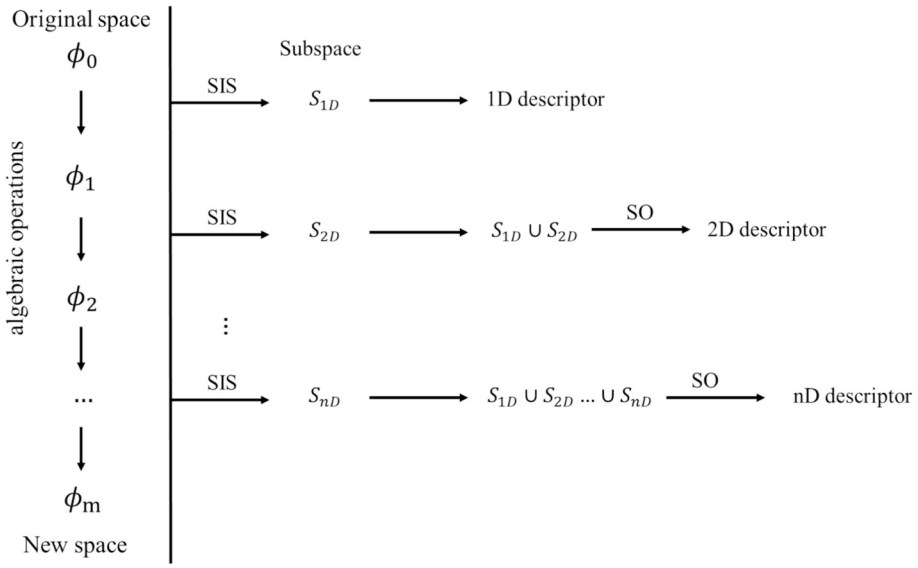


Fig. 3. Schematic diagram of the SISSO algorithm.

Table 3
Definition of two design spaces.

Design space	Imperfection ratio	Layup scheme	Material
Design space 1	[0.1%, 1%]	$[90_4/(\pm\alpha/\pm\beta)_8/90_4]$	T700/Epoxy
Design space 2	0.8%	$[90_4/(\pm\alpha/\pm\beta)_8/90_4]$	T800G/5228
		$[90_4/(\pm\alpha/\pm\beta)_8/90_4]$	M40J/TDE-85
		$[(\pm\alpha/\pm\beta)_{10}]$	T700/Epoxy

Table 4
Comparison of our FEA results and the experiments.

Case	Experiment (MPa)	Our result (MPa)	Relative error (%)
Chen et al.[6]	8.04	7.630	5.10
Jiang et al.[11]	72	72.770	1.07

3.2. *Influence of design parameters on failure mode and UTS of composite hulls*

The influence of various design parameters on failure mode and UTS is explored in this section. Unless otherwise specified, the following baseline parameters are used: T700/Epoxy material, ply angles of 50° and 80°, a thickness of 17 mm, a length of 600 mm, a radius of 100 mm, and an imperfection ratio of 0.1%. Fig. 5 presents the variations in failure modes with respect to the winding angles (α, β), as well as the radius R and thickness t. In Fig. 5(a), both α and β vary from 30° to 80° in 5° increments. In Fig. 5(b), the radius R ranges from 80 mm to 160 mm in 10 mm steps, while the thickness t varies from 12 mm to 20 mm in 1 mm increments. The FEA results show that design parameters significantly affect the failure modes. Green and yellow square markers represent samples with failure modes of PBIPF and MDDF, respectively. For example, a lower winding angle promotes the PBIPF failure mode, whereas a lower R/t ratio leads to MDDF, which is consistent with the findings in reference [12]. Fig. 5 also illustrates that PBIPF constitutes majority of failure cases, with MDDF representing only a small subset in the current design space. Additionally, the colored spherical markers in Fig. 5 serve as test samples for validating the SISSO-based classification model, which will be discussed in subsequent sections.

Fig. 6 depicts the variations of UTS with thickness t, radius R and imperfection ratio Ir for composite hulls. Due to the predominance of PBIPF cases, only samples exhibiting PBIPF are considered here. All samples in Fig. 6 are based on a T700/Epoxy material with a layup scheme of $[90_4/(\pm20/\pm70)_8/90_4]$, and have a fixed length of 400 mm. The baseline geometry is defined by a thickness of 5 mm, a radius of 100 mm, and an imperfection ratio of 0.8%, except for the parameters being varied. In Fig. 6(a), the thickness t varies from 5 mm to 20 mm while the radius R ranges from 100 mm to 180 mm. Fig. 6(b) shows the effect of the imperfection ratio Ir, which varies from 0.1% to 1.0%. The results reveal a highly nonlinear relationship between these design parameters and the UTS. Specifically, Fig. 6(a) reveals that greater thickness and smaller radius lead to higher UTS. By adjusting geometric parameters, the UTS increased from 3 MPa to 120 MPa, indicating that geometric parameters exert a stronger influence on hull's strength. An increased imperfection ratio significantly reduces UTS, as shown in Fig. 6(b). The UTS value decreases by 18% as Ir increased from 0.1% to 1%. The variation curves of load with ARC length for samples with Ir of 0.1%, 0.4% and 0.8% demonstrate that structures with different Ir exhibit similar failure behaviour. These findings highlight the complex interplay between geometric and ply parameters in governing hull's performance. To further examine the influence of imperfection shape, a preliminary sensitivity study is conducted by considering several higher modes and their combinations in Fig. S3, Fig. S4 and Table S2. These results indicate that higher-order and mixed-mode imperfections can influence the ultimate strength, while first mode provides a conservative estimate for the representative cases examined. Consequently, more advanced methods are needed to explore an expanded design space that includes ply angle, thickness, radius, material, and imperfection ratio.

4. Predictions of ultimate failure mode and strength by SISSO

4.1. *Feature engineering*

In order to make the proposed approach more universal, stiffness coefficients, i.e. A_{ij}, B_{ij}, and D_{ij} of laminate are employed, allowing the approach to be applied across a broader range of materials and layup schemes [7]. Specifically, A_{ij} represent the extensional stiffness coefficients, B_{ij} the coupling stiffness coefficients, and D_{ij} the bending stiffness coefficients as shown in Eqs. (S11-S13). The design parameters of composite hulls involve 32 features, including length L, thickness t, radius R, layup angle (α, β), stiffness coefficients $(A_{11}, A_{12}, A_{22}, A_{33}, B_{11}, B_{12}, B_{13}, B_{22}, B_{23}, B_{33}, D_{11}, D_{12}, D_{13}, D_{22}, D_{23}, D_{33})$, material properties $(E_1, E_2, \nu_{12}, G_{12}, X_t, X_c, Y_t, Y_c, S)$, imperfection ratio Ir. Besides, eigenvalue Q, obtained from efficient FEA simulations, is also incorporated as an input feature due to its proven enhancement of predictive accuracy.

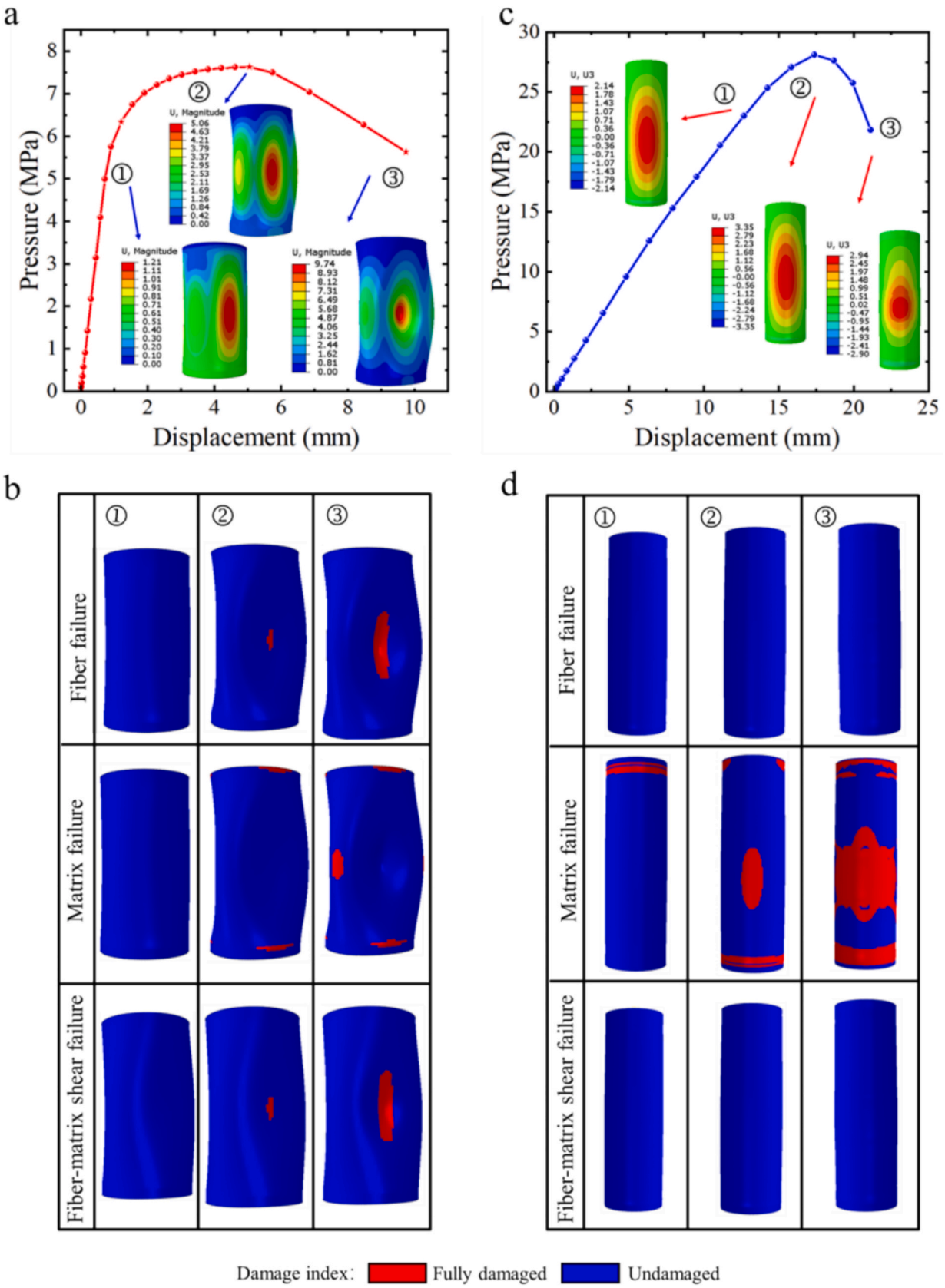


Fig. 4. Variation curves of load and on composite cylindrical shells and the damage distribution of the FE model for the sample with PBIPF (a-b) and sample with MDDF (c-d).

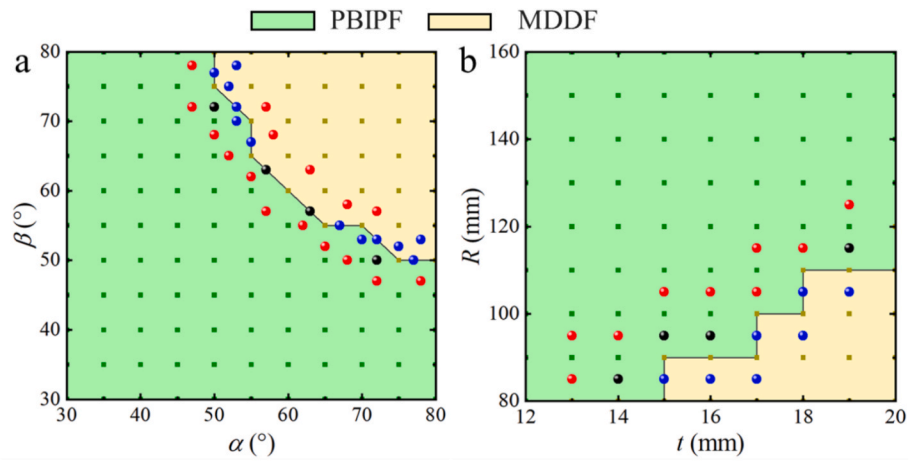


Fig. 5. Variations of failure modes for composite hulls with $[90_4/(\pm\alpha/\pm\beta)_8/90_4]$ under different design parameters: angle α versus angle β (a), and radius R versus thickness t (b). The green and yellow square markers represent FEA samples. The colored spherical markers are predictions by the SISSO classification model, which will be discussed later.

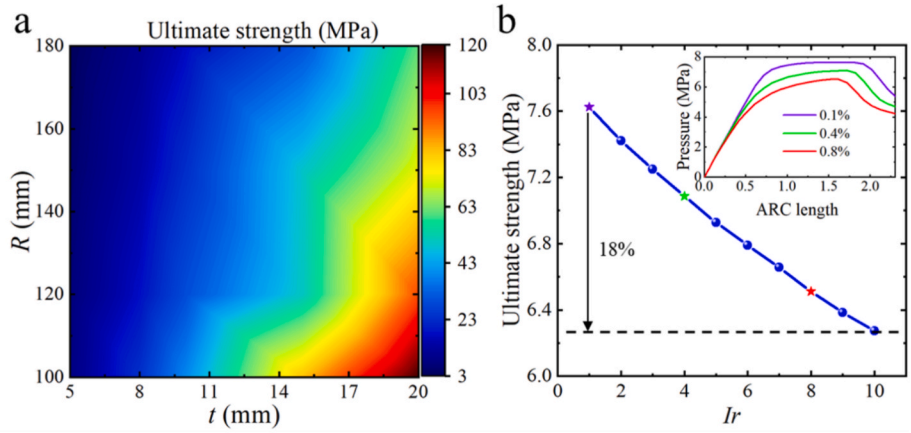


Fig. 6. Variations of UTS for composite hulls with different design parameters: thickness t versus radius R (a), and imperfection ratio Ir (b).

However, training a high-dimensional model with all features is computationally expensive and prone to overfitting [39,40]. To address this, systematic feature selection is performed to eliminate redundant features. Correlation and importance analysis is employed to assess influence of each feature on the UTS and inter-correlations among features, as shown in Fig. S5. The analysis reveals complex nonlinear relationships between design parameters and UTS which is denoted as P. Among them, Q exhibits the strongest positive correlation with P. Stiffness coefficients, which are functions of material properties and lamination angles, comprehensively represent these underlying parameters. Among the original variables, thickness t and bending stiffness component D_{22} (primarily controlled by thickness and stacking sequence) exhibit the strong positive correlations with UTS. This suggests that, within the considered design space, variations in laminate stiffness have a more pronounced influence on strength than global geometric dimensions. Although radius R and imperfection ratio Ir show weak correlations with other features in Fig. S5(a), they exert a significant negative influence on structural strength, as evidenced in Fig. 6. Consequently, we selected eight highly influential features $(Q, R, t, A_{22}, B_{23}, D_{12}, D_{22}, D_{23})$ from Fig. S5(b), along with two relatively independent yet physically critical features (Ir, L), to train the SISSO model.

4.2. Classification of failure modes

The performance of SISSO-based classification model in predicting the failure modes of composite hulls is discussed here, which are categorized as either PBIPF or MDDF. The design space 1 is selected as the physical space. A total of 1000 samples are randomly collected through FEA. Among them, 167 samples are identified as MDDF, while the remaining cases correspond to PBIPF. The determined ten features are employed to construct the key descriptors, denoted as: $Des_1 = \frac{\sqrt[3]{Q}(A_{22}+B_{23})B_{23}}{D_{33}\ln(D_{22})}$, $Des_2 = \left(\frac{L}{R}+\frac{D_{12}}{B_{23}}\right)\frac{\sqrt{D_{33}}}{t^3}$, $Des_3 = \frac{(D_{33}-D_{12})}{\sqrt{D_{33}}(\frac{L}{R}+\frac{D_{12}}{B_{23}})}$. The classification results are summarized in Fig. 7 and Table S2. It can be found from Fig. 7 that the three descriptors obtained by SISSO effectively reduce the dimensionality of the input features and separate the distribution space of the two failure modes. Although some overlap exists between the two regions, only a small proportion of boundary samples are misclassified. As shown in Table S2, the classification model achieved accuracy, precision, and recall exceeding 90%, validating its reliability in recognizing failure modes. Additionally, Des_1 serves as the primary discriminant: when $Des_1 >$ -4000, most samples correspond to PBIPF, whereas when $Des_1 <$ -6000, the majority correspond to MDDF. Within the range of Des_1 between [-6000, −4000], both Des_2 and Des_3 are required for accurate discrimination. Specifically, a high value of Des_2 combined with a low value of Des_3 indicates a higher likelihood of MDDF. Among the descriptors, Des_1 is primarily stiffness-related, as it comprises stiffness coefficients and the eigenvalue Q, while Des_2 and Des_3 represent coupling terms associated with geometry and stiffness. This confirms that structural stiffness is the key factor governing failure

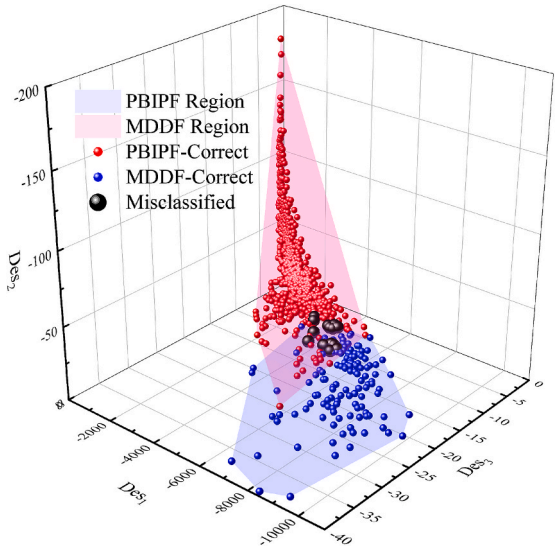


Fig. 7. Distribution of classification results for different failure modes based on SISSO.

modes.

Furthermore, Fig. 5 provides an intuitive interpretation by correlating failure modes with specific design variables, clarifying the role of design parameters in determining structural failure behavior. Additional test samples are generated near the boundary between the two failure modes for model evaluation. In Fig. 5, red spherical markers represent samples correctly classified as PBIPF, blue ones represent MDDF, and black markers indicate misclassified samples. The distribution demonstrates a high accuracy of the SISSO-predicted failure modes, as majority of samples are correctly grouped.

We perform a global sensitivity analysis using SHAP (SHapley Additive exPlanations) on the classification model, as shown in Fig. S6. SHAP quantifies the marginal contribution of each feature to the model output by measuring its impact on the predicted classification. The mean absolute SHAP values are used to rank feature importance, with larger absolute values indicating greater sensitivity and stronger contributions to the model predictions. It is observed that the three SISSO descriptors exhibit significantly higher sensitivity than any individual physical variable, demonstrating that the SISSO framework effectively condenses high-dimensional physical information into a few dominant governing features. Notably, Des_1, identified as the most important descriptor in the SHAP analysis, is primarily stiffness-related. This observation is consistent with the feature-level SHAP results, where A_{22} and B_{23} show the highest contributions among the original variables, suggesting that their influence is effectively captured and amplified within Des_1. In contrast, geometric parameters L and R contribute weakly. This indicates that the failure mode transition is primarily governed by the balance between in-plane and bending stiffness characteristics, rather than purely geometric scaling effects, which is consistent with conclusions reported in the literature[41].

4.3. Prediction of ultimate strength

4.3.1. SISSO regression model for composite hulls with MDDF

This section discusses the prediction of UTS for composite hulls exhibiting MDDF using SISSO regression model. The dataset consists of 167 samples with MDDF from the design space 1. Of these, 80% samples are randomly selected as the training set, while the remaining 20% constitute the testing set to evaluate prediction performance. The formula obtained by SISSO is presented as follows:

$$P_{MDDF} = c_1\frac{(D_{33}-D_{12})D_{33}}{B_{23}(t+R)(A_{22}+D_{22})} + c_2\frac{\sqrt[3]{IrL}}{B_{23}}\left(\frac{D_{33}}{B_{23}}+\frac{D_{22}}{D_{33}}\right) + c_0 \quad (1)$$

where c_1, c_2, and c_0 are constants. Specifically, $c_1 = -16.2, c_2 = -6$, $c_0 = 37.8$. The predictions for the testing set are presented in Fig. 8. For composite hulls with MDDF, the UTS values predicted by Eq. (1) closely match the reference results from FEA. The calculated RMSE is 3.162 and R^2 is 0.962, demonstrating its reliability in predicting UTS for this failure mode. Furthermore, the equation's structure, which incorporates geometric and stiffness-related descriptors but lacks the eigenvalue Q, proves that this failure mode is governed more by material failure than by buckling instability. Despite the limited training data, the explicit physical interpretation of Eq. (1) remains somewhat ambiguous. Nevertheless, it retains significant value for the direct prediction of target properties in practical engineering.

4.3.2. SISSO regression model for composite hulls with PBIPF

The performance of SISSO regression model in predicting UTS of composite hulls with PBIPF is discussed in this section. The model is constructed using data from the two design spaces outlined in Table 3, ensuring coverage of key design variables across geometry, material, and lay-up. A total of 4000 samples without MDDF are generated: design space 1 contains 1000 samples, while design space 2 comprises 1000 samples for each of the two materials and 1000 samples for new layup scheme. 80% of samples are randomly selected to train the model, and the rest 20% are reserved as the testing set to validate prediction accuracy. The resulting formula is expressed as follows:

$$P_{PBIPF} = d_1 lgQ\frac{t}{R} + d_2 Q\frac{D_{22}}{(D_{33}-D_{12})} + d_3 Q\frac{IrL}{A_{22}} + d_0 \quad (2)$$

where d_1, d_2, d_3 and d_0 are constants. Specifically, $d_1 = 20.3$, $d_2 = 10.3$, $d_3 = -5346$, $d_0 = 32$. The prediction results of testing data are shown in Fig. 9. Overall, the regression equation achieves satisfactory performance, with the RMSE and R^2 calculated as 5.04 and 0.961, respectively. The physical interpretation of Eq. (2) is further elucidated: the first term corresponds to geometric effects, the second to stiffness-related effects, and the third to their coupled interaction. It is noteworthy that the eigenvalue Q is embedded in all three terms, emphasizing its significant correlation with UTS—a result that aligns with the correlation analysis in Fig. S5 and supports viewing the derived expression as a modified eigenvalue-based formulation. In addition, the positive value of coefficient d_1 indicates that the UTS is negatively correlated with R/t, consistent with the findings reported in [12]. Meanwhile, the negative value of d_3 suggests that both L and Ir are negatively correlated with the UTS, consistent with the conclusions in Section 4.1.

Eq. (2) is further validated for its predictive performance outside the

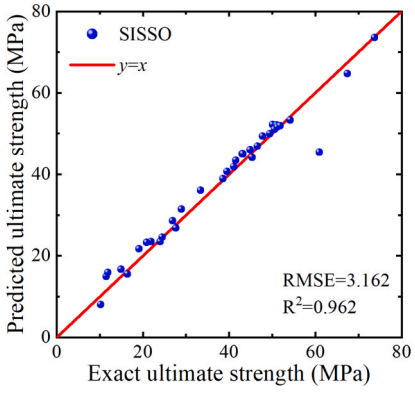


Fig. 8. Prediction results of UTS of composite hulls with MDDF based on SISSO regression model.

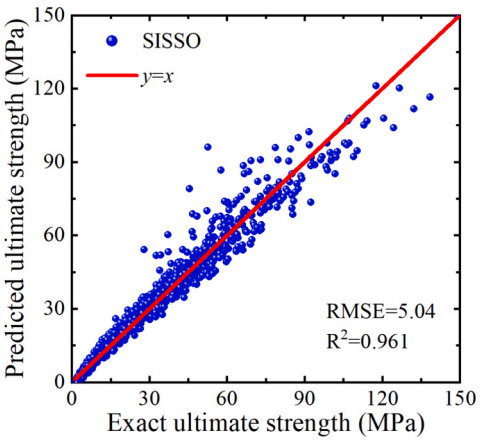


Fig. 9. Prediction results of UTS of composite hulls with PBIPF based on SISSO regression model.

design space through two approaches: published experimental cases and a newly generated dataset. Firstly, published experimental cases are collected, as shown in Table S4. Note that the layup schemes adopted in references [6] and [11]—$[90_4/(90/\pm45/0)_8/90_4]$ and $[(90_2/\pm20/90_2/\pm45/90_2/\pm70)_5/90_4]$, respectively—differ from those in the present work. The materials in [11,37] and [42], namely T700 composite, UD600-SGL50 K CFRP composite and glass/epoxy, also fall outside the defined material design space. The UTSs obtained from experiments and Eq. (2) are compared in Table 5 and Fig. 10(a). Secondly, a new test set of 800 PBIPF samples is generated using the new material of IM7-8552 from [43], while keeping other design parameters consistent with design space 2. Additionally, two empirical formulas as defined in Eqs. (S14-S15) , NASA SP-8700 and ASME Code 2007(RD-1172), are also used to predict the UTS of cases in experiments and the newly collected 800 samples.

As shown in Table 5 and Fig. 10(a), Eq. (2) demonstrates superior predictive capability for the experimental cases, with the discrepancies between the predicted and experimental values all within 13%. This robustness stems from the incorporation of stiffness coefficients and the eigenvalue Q, which inherently depend on material properties and layup schemes, thereby granting the model a degree of scalability. Furthermore, the consistent accuracy observed across different composite systems indicates that the SISSO-derived descriptors possess good generalization capability. This can be attributed to the stiffness-based representation, which inherently captures the combined effects of material properties, fiber architecture, and stacking sequence. In contrast, there are some fluctuations for the prediction accuracy of the NASA and ASME formulas. A similar trend is observed for the 800 new samples in Fig. 10(b). The NASA and ASME formulas are primarily applicable to samples with low t/L and t/R ratios and tend to overestimate UTS for other specimens due to their neglect of progressive damage effects [16]. Meanwhile, the relative errors of Eq. (2) are predominantly distributed within ± 20%, achieving an RMSE of 6.9 and an R^2 of 0.95. Therefore, it can be concluded that the proposed model delivers accurate predictions across the broad parameter space, confirming its robustness and generalizability for predicting the UTS of composite hulls with PBIPF.

Table 5
Comparison of experiments and predicted UTSs of composite hulls obtained from NASA, ASME and our work. (MPa for UTS, % for relative error, (RE)).

Experiment	Our	RE	NASA's	RE	ASME's	RE
8.04 [6]	8.75	8.79	11.139	38.54	8.41	4.64
72 [11]	65.98	8.36	80.71	12.10	128.82	78.92
4.86 [37]	4.86	0.33	3.69	24.33	2.59	46.93
3.30 [37]	3.01	8.85	2.23	32.38	1.66	49.62
2.11 [37]	2.12	0.62	1.58	24.86	1.23	41.89
15.54[41]	13.99	10.00	23.55	51.53	26.59	71.10
13.38[41]	13.15	3.03	21.00	56.97	24.17	80.67
11.82[41]	12.41	1.21	19.25	62.89	22.17	87.54
14.89[41]	16.09	4.69	11.22	24.68	8.75	41.26
13.41[41]	15.05	10.23	10.20	23.95	7.26	45.83
12.93[41]	14.59	12.85	9.19	28.93	6.66	48.52
16.4[42]	16.17	1.38	12.32	24.87	12.72	22.42

4.4. Direct design of composite hulls based on the UTS formula

The SISSO-based predictive framework establishes a foundation for inverse design in composite hull structures. By generating equivalent strength surfaces, it directly links target performance (UTS) with critical design variables—thickness and layup angles. Given the limited number of MDDF samples, only Eq. (2) is discussed here. Assuming a composite hull is made of T700/Epoxy, with a length of 800 mm, radius of 50 mm, imperfection ratio of 0.5%, and layup scheme $[90_4/(\pm\alpha/\pm\beta)_8/90_4]$. Using Eq. (2), equivalent strength surfaces for various UTS values are constructed in the 3D space of thickness (t) and layup angles (α, β), as shown in Fig. 11. These surfaces provide an intuitive approach to performance-based design.

For instance, with a thickness of 10 mm and layup angles of $\alpha = 10^\circ$ and $\beta = 30^\circ$, the composite hull achieves a UTS of 30 MPa. Increasing the thickness to 12 mm while keeping the layup angles unchanged raises the UTS to 40 MPa ($P_1 \rightarrow P_2$). Equivalent thickness surface (the light blue flat plane) further allows structural performance improvements through adjustments of ply angles without increasing hull's thickness. For example, maintaining a thickness of 12 mm, the UTS can be increased to 50 MPa by adjusting the layup angles from 10° and 30° to 40° and 70° ($P_2 \rightarrow P_3$). Additionally, by optimizing the ply angles on a given equivalent strength surface, laminate thickness can be reduced for weight-saving designs without compromising strength. For a sample with a thickness of 13 mm and layup angles of 10° and 10°, its thickness can be reduced to 12 mm by adjusting its angle to 10° and 30°, whilst maintaining its UTS at 50 MPa ($P_4 \rightarrow P_3$), effectively reducing material usage. Consequently, the SISSO-derived formula serves as an efficient and robust tool for guiding the structural design and long-term performance optimization of composite hulls, eliminating the need for repetitive, computationally expensive simulations.

5. Conclusion

In this work, an interpretable data-driven approach is developed to derive explicit formulas for predicting the UTS and failure mode of composite hulls across a wide range of design parameters. Experimentally calibrated FEA is first conducted to investigate the ultimate failure behavior of composite hulls under various design parameters. Two primary failure modes are identified: PBIPF, characterized by the coupling of fiber failure and buckling deformation, and MDDF, manifested by extensive matrix failure with slight buckling deformation. Parametric studies reveal a highly nonlinear relationship between the design parameters and both failure modes and UTSs. Leveraging the FEA data, a SISSO model is employed to derive explicit formulas to predict both failure modes and UTSs. The reliability and generalizability of the derived expressions are systematically validated through comparisons with published experimental results and existing empirical formulas (ASME 2007 and NASA SP-8700). Results show that the SISSO-based classification and regression models demonstrate high predictive accuracy and physical interpretability. Specifically, the classification expression identifies structural stiffness as the dominant factor governing failure mode transitions. The regression formula for PBIPF—integrating structure-related, stiffness-related, and coupling terms—enables direct visualization of the influence of each design variable on the structural performance. Moreover, the explicit SISSO-derived equations provide a straightforward and efficient tool for the design and optimization of composite hulls. The framework thus provides a practical, data-enhanced tool for the robust design and failure evaluation of

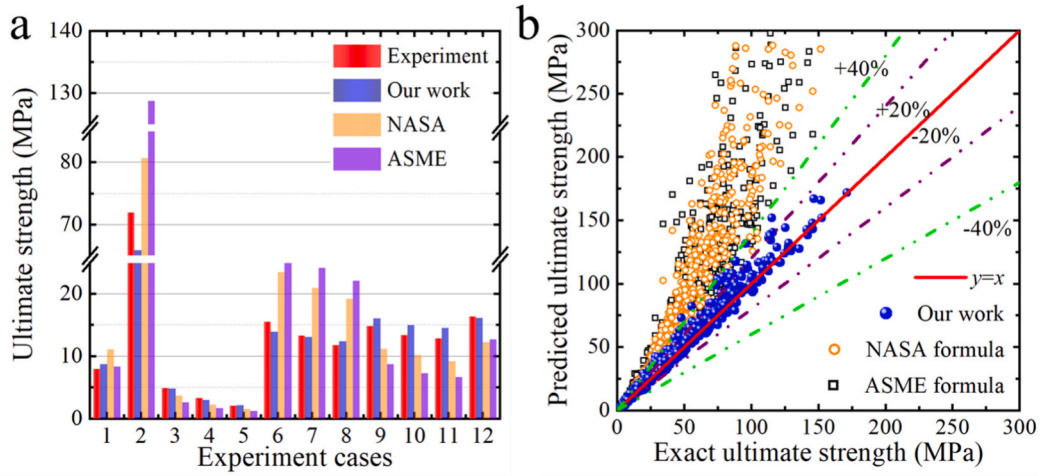


Fig. 10. Comparison of experiments (a) and predicted UTSs of composite hulls obtained from NASA, ASME and our work (b).

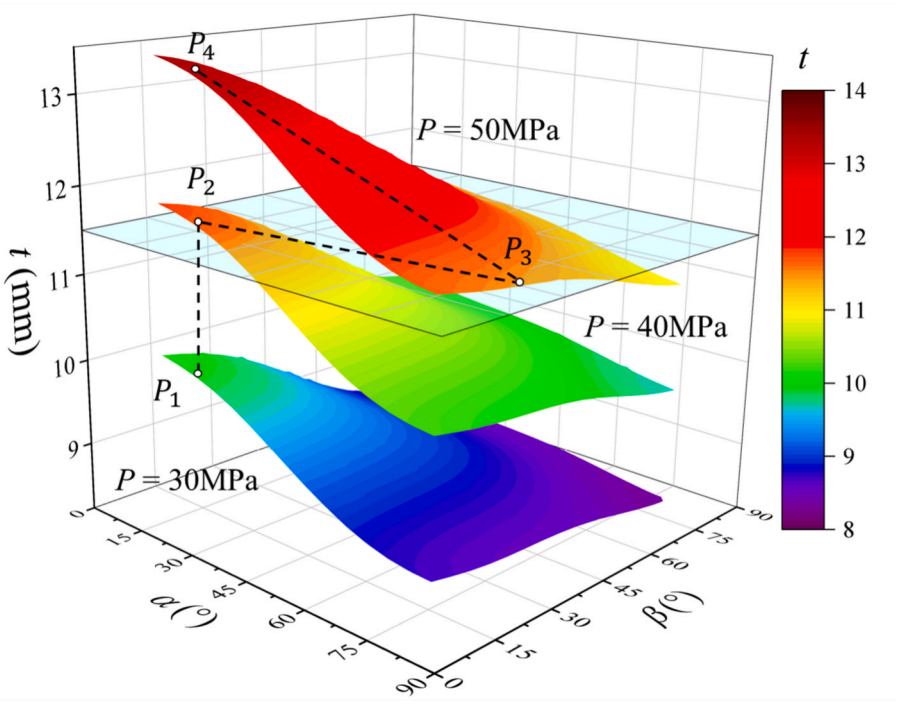


Fig. 11. Illustration of direct design strategies based on the UTS space constructed by the SISSO formula.

composite hulls. It should be noted that, in cases with defects characterized by higher-order or mixed-mode patterns, the current study may exhibit limitations in predicting their UTS and failure mode. Future research will focus on incorporating these factors to further enhance the versatility and robustness of the proposed formula.

CRediT authorship contribution statement

Qingfeng Wang: Writing – original draft, Validation, Software, Methodology, Data curation. **Ziyi Li:** Visualization, Software, Methodology. **Zhengpen Liu:** Software, Methodology. **Shu Lin:** Investigation. **Liyong Jia:** Software, Formal analysis. **Yushu Li:** Writing – review & editing, Visualization, Validation, Methodology, Formal analysis. **Yilun Liu:** Visualization, Supervision, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors acknowledge the financial support of National Natural Science Foundation of China (12325204, 12502134), Shaanxi Province Natural Science Basic Research Program (2025JC-YBMS-072), China Postdoctoral Science Foundation (2024 M762574), and Fundamental Research Funds for the Central Universities (xzy012025039).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.

org/10.1016/j.compositesa.2026.109863.

Data availability

Data will be made available on request.

References

[1] Air A, Gangadhara Prusty B. Manufacturing feasibility of a bend free ellipsoidal composite pressure vessel using automated fibre placement. Compos A Appl Sci Manuf 2024;177:107968.
[2] Wu Q, Liu X, Li J, Li J, Wei X, Zhao Y, et al. Failure of carbon fiber composite sandwich cylinders with a lattice core under axial compressive loading. Compos A Appl Sci Manuf 2022;155:106812.
[3] Zuo X, Zhang J, Tang W, Li Y, Li H. Buckling behavior of steel and steel–composite cylinders under external pressure. Thin-Walled Struct 2022;181:110011.
[4] Almeida JHS, Ribeiro ML, Tita V, Amico SC. Damage and failure in carbon/epoxy filament wound composite tubes under external pressure: Experimental and numerical approaches. Mater Des 2016;96:431–8.
[5] Wang W, Zhu Y, Zhou L, Zhang J. Buckling characteristics of carbon fiber composite/aluminium alloy combined cylindrical shells under external hydrostatic pressure. Int J Press Vessel Pip 2024;208:105125.
[6] Chen M, Xinhu Z, Pan G. Data-driven approach for uncertainty quantification and risk analysis of composite cylindrical shells for underwater vehicles. Mech Adv Mater Struct 2024;31(17):4116–30.
[7] Wei R, Pan G, Jiang J, Shen K, Lyu D. An efficient approach for stacking sequence optimization of symmetrical laminated composite cylindrical shells based on a genetic algorithm. Thin-Walled Struct 2019;142:160–70.
[8] Guan W, Zhu Y, Wang W, Wang F, Zhang J, Wu Y, et al. Experimental and numerical buckling analysis of carbon fiber composite cylindrical shells under external pressure. Ocean Eng 2023;275:114134.
[9] Imran M, Shi D, Tong L, Elahi A, Waqas HM, Uddin M. Multi-objective design optimization of composite submerged cylindrical pressure hull for minimum buoyancy factor and maximum buckling load capacity. Def Technol 2021;17(4):1190–206.
[10] Shen K-C, Pan G. Buckling and strain response of filament winding composite cylindrical shell subjected to hydrostatic pressure: Numerical solution and experiment. Compos Struct 2021;276:114534.
[11] Jiang X, Yu C, Li Y, Zeng Y. Ultimate state assessment of medium-thick composite cylindrical pressure shells based on progressive damage. Ocean Eng 2023;289:116128.
[12] Cong F, Zhang R, Li W, Jin Y, Yu G, Wu L. Buckling analysis of moderately thick carbon fiber composite cylindrical shells under hydrostatic pressure. Appl Ocean Res 2024;153:104272.
[13] Zhang X, Liu Y, Chen Z, Yang Z, Hu R, Pan G. Buckling and material failure analyses for ring-stiffened composite shells subjected to hydrostatic pressure. Appl Ocean Res 2024;146:103944.
[14] Yuan K, Xia Z, Wang L, Wu L. Mechanical properties and failure analysis of ring-stiffened composite hulls under hydrostatic pressure. Compos Struct 2025;351:118609.
[15] Zhang X, Li Z, Wang P, Pan G. Experimental and numerical analyses on buckling and strength failure of composite cylindrical shells under hydrostatic pressure. Ocean Eng 2022;249:110871.
[16] Cho YS, Oh DH, Paik JK. An empirical formula for predicting the collapse strength of composite cylindrical-shell structures under external pressure loads. Ocean Eng 2019;172:191–8.
[17] Wei R, Pan G, Jiang J, Shen K. Influence of ply angle and length on buckling behavior of composite shells under hydrostatic pressure. J Reinf Plast Compos 2019;38(10):478–91.
[18] Zhang X, Li Z, Yang Z, Jiang L, Pan G. Buckling of composite shells with a novel initial imperfection model subjected to hydrostatic pressure. Compos Struct 2022;297:115949.
[19] Hong H, Kim W, Kim S, Lee K, Kim SS. Deep transfer learning for efficient and accurate prediction of composite pressure vessel behaviors. Compos A Appl Sci Manuf 2024;186:108413.
[20] Portales Picazo P, Gray A, Zobeiry N. Efficient characterization and optimization of pyrolysis in carbon-carbon composites through machine learning. Compos A Appl Sci Manuf 2025;190:108664.
[21] Z.u.R. Tahir, P. Mandal, M.T. Adil, F. Naz, Application of artificial neural network to predict buckling load of thin cylindrical shells under axial compression, Eng. Struct. 248 (2021) 113221.
[22] Liu J, He B, Yan T, Yu F, Shen Y. Study on carbon fiber composite hull for AUV based on response surface model and experiments. Ocean Eng 2021;239:109850.
[23] Liu R, Lin S, Wan J, Li L, Zhang G, Qin H, et al. Recent advances in machine learning guided mechanical properties prediction and design of two-dimensional materials. Thin-Walled Struct 2025;213:113261.
[24] Zong B, Li J, Yuan T, Wang J, Yuan R. Recent progress on machine learning with limited materials data: using tools from data science and domain knowledge. J Materiomics 2025;11(3):100916.
[25] Zhang Z, Zhang Z, Di Caprio F, Gu GX. Machine learning for accelerating the design process of double-double composite structures. Compos Struct 2022;285:115233.
[26] Okafor CE, Iweriolor S, Ani OI, Ahmad S, Mehfuz S, Ekwueme GO, et al. Advances in machine learning-aided design of reinforced polymer composite and hybrid material systems. Hybrid Adv 2023;2:100026.
[27] Hasebe S, Higuchi R, Yokozeki T, Takeda S-I. Prediction of compression after impact strength from surface profile of low-velocity impact damaged CFRP laminates using machine learning. Compos A Appl Sci Manuf 2025;189:108560.
[28] Jenkins W, Gross H, Detwiler K, Morscher G, Przybyla C, Hochhalter J. Developing a continuum damage model for ceramic matrix composites using genetic programming based symbolic regression. Compos Struct 2025;369:119274.
[29] Zou D, Thoti D, Bao Z. Study on the application of discrepancy-guided symbolic regression algorithm in analyzing the impact resistance of UHP-SFRC target against high velocity projectile impact. Int J Impact Eng 2025;201:105276.
[30] Dong J, Zhong J. Recent advances in Symbolic Regression. ACM Comput Surv 2025;57(11). Article 290.
[31] Ouyang R, Curtarolo S, Ahmetcik E, Scheffler M, Ghiringhelli LM. SISSO: a compressed-sensing method for identifying the best low-dimensional descriptor in an immensity of offered candidates. Phys Rev Mater 2018;2(8):083802.
[32] Tian S, Zhou K, Yin W, Liu Y. Machine learning enables the discovery of 2D Invar and anti-Invar monolayers. Nat Commun 2024;15(1):6977.
[33] Pan R, Tu X, Ma X, Liu L, Yan T, Tong M. Interpretable machine learning on C3H6 and C3H8 diffusion in covalent organic frameworks: Incorporating the effects of framework flexibility. Chem Eng Sci 2025;310:121520.
[34] Chen F, Guo Z, Wang J, Ouyang R, Ma D, Gao P, et al. Accelerated feasible screening of flame-retardant polymeric composites using data-driven multi-objective optimization. Comput Mater Sci 2023;230:112479.
[35] Lee D-K, Nam K-W, Kim W-J, Park S-H. Development of hybrid composites with zero-temperature coefficient of resistance: Mechanisms and predictive modeling using symbolic regression. Chem Eng J 2025;506:159866.
[36] Fu X, Mei Z, Li H, Chen G, Wang S, Zhang E, et al. Ultimate bearing behavior in the post-buckling stage of composite pressure hulls based on new detection methods. Structures 2024;63:106413.
[37] Zuo X, Jian Z, Wenxian T, Ming Z, Li Y. Buckling of helically wound composite cylinders under uniform external pressure. Ships Offshore Struct 2024;19(3):348–65.
[38] Pan ZZ, Zhang LW, Liew KM. The use of curvilinear fibers for enhancement of progressive failure performance of perforated composite panels. Compos Struct 2022;288:115424.
[39] Xu Y, Qian Q. i-SISSO: Mutual information-based improved sure independent screening and sparsifying operator algorithm. Eng Appl Artif Intel 2022;116:105442.
[40] Wang X, Cao Y, Ji J, Sheng Y, Yang J, Ke X. A multi-objective, multi-interpretable machine learning demonstration verified by domain knowledge for ductile thermoelectric materials. J Materiomics 2025;11(2):100886.
[41] Hao W, Ming Z, Yongsheng L, Lihui W. Explainable machine-learning-assisted failure analysis of moderately thick composite cylindrical shells under hydrostatic pressure. Int J Press Vessel Pip 2026;219:105672.
[42] Li Y, Wang W, Li H, Qu P, Jian Z. Ultimate strength of imperfect moderate thick composite cylindrical pressure shell under hydrostatic pressure. Acta Materiae Compositae Sinica 2023;40(5):2639–52.
[43] Li Y, Qin H, Tan VBC, Jia L, Liu Y. A deep transfer learning approach to construct the allowable load space of notched composite laminates. Compos Sci Technol 2024;247:110432.